

Data Dictionary

Purpose:

The Data Dictionary provides detailed documentation of the fields (columns) across the synthetic dataset. It also identifies data generation issues and describes how each was fixed or improved to enhance realism, accuracy, and usability in analysis.

No. Field / Issue		Description / Fix Applied
1	Flat gender / customer_type	Added weighted probabilities for more realistic gender and customer_type balance
2	Age random between 18–70	Applied a skewed normal distribution, concentrating around common traveling ages (25–40)
3	All emails perfectly generated	Introduced missing values and typos to simulate real-world inconsistencies
4	Names may not match gender	Assigned gender-specific names to maintain consistency
5	No duplicates or noise	Injected optional dirty rows and random noise (e.g., weird characters or blanks)
6	package_name	Ensured no duplicates — each destination is used only once
7	package_type	Made type relevant to destination (e.g., beachy places = honeymoon)
8	price_per_pax	Balanced prices across realistic travel tiers (budget, mid, premium)
9	duration_days	Longer durations for premium/far destinations
10	category	Made category align better with type/destination (e.g., hiking = adventure)
11	role	Used logic to assign fewer main_guides, more assistant_guides
12	working_status	Used weighted probabilities — “permanent” is most common
13	commission_rate	Kept existing logic (already realistic)
14	booking_date	Distributed bookings based on seasonality (e.g., school holidays = peak)

No. Field / Issue	Description / Fix Applied
15 travel_date	Always occurs after booking (1–90 day window)
16 discount_promotion	Skewed distribution — most users get 0–5%, only a few get high discounts
17 payment_status	Tied to travel_date — if in future = pending
18 payment_method	Weighted common methods higher — credit card & online dominate
19 SatisfactionScore	Skewed towards positive but with variance
20 Rating_guide, hotel, itinerary	Based on satisfaction ± small random variation
21 Comments	Tone matches sentiment of score (positive = good tone)
22 WouldRecommend	Already solid — left as-is
23 refund_rows	Only included bookings where payment_status = 'refunded'
24 refund_amount	Calculated as price × pax × refund %
25 refund_percentage	Tied to refund reason (e.g., emergency = higher %)
26 refund_date	Occurs after booking but before travel (or shortly after travel)
27 refund_status	Based on timing or randomness (e.g., approved = early + good reason)